Research Methodology in Public Administration

Proposal Writing and Report Writing

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Sampling Techniques

Selecting a Sample from a Population

Population

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Sampling Technique

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Sample

What is Sampling?

Sampling is the process of selecting a smaller group of units from a larger population for the purpose of conducting a research study.

Instead of studying every member of the population, the researcher studies a selected sample.

Example

If a researcher wants to study all students of Tribhuvan University, it may not be possible to collect data from every student.
Therefore, a group of students is selected as a **sample**.

Basic Concepts in Sampling

Population: The complete group that the researcher wants to study.

Target Population: The specific population to which the researcher wants to generalize the findings.

Sample: A smaller group selected from the population.

Sampling Unit: The individual element or unit selected for the study.

Sampling Frame: A list or source from which the sample is selected.

Why Do We Use Sampling?

Sampling is used because studying the entire population may be:

- Expensive
- Time-consuming
- Difficult to manage
- Impractical

A properly selected sample can provide useful information about the larger population.

Major Types of Sampling

Sampling techniques are broadly divided into two categories:

Probability Sampling

Every unit in the population has a known probability of being selected.

- Simple Random
- Systematic
- Stratified
- Cluster

Non-Probability Sampling

The probability of selecting each unit is unknown.

- Convenience
- Purposive
- Quota
- Snowball

Probability Sampling

In probability sampling, every member of the population has a **known and non-zero probability** of being selected.

It generally involves some form of random selection.

Major Advantage

Probability sampling allows the researcher to make statistical inferences from the sample to the larger population.

1. Simple Random Sampling

In **simple random sampling**, every member of the population has an equal chance of being selected.

Example:
Suppose there are 1,000 students in a university.

The researcher wants to select 100 students.

Each student is assigned a number, and 100 numbers are selected randomly.

Methods

Lottery method	Random number table	Computer-generated random selection

Advantages and Limitations of Simple Random Sampling

Advantages

- Simple to understand
- Reduces selection bias
- Supports statistical analysis

Limitations

- Requires a complete sampling frame
- May be difficult for large populations
- Can be costly and time-consuming

2. Systematic Sampling

In **systematic sampling**, units are selected at regular intervals from a sampling frame.

Sampling Interval

$$k = \frac{N}{n}$$

Where:

- N = Total population
- n = Required sample size
- k = Sampling interval

Example: Systematic Sampling

Suppose:

- Population $=$ 1,000 students
- Required sample $=$ 100 students

Therefore:

$$k = \frac{1000}{100} = 10$$

The researcher randomly selects a starting point and then selects every **10th student**.

Example:

$$3,\ 13,\ 23,\ 33,\ 43, \ldots$$

3. Stratified Random Sampling

In **stratified sampling**, the population is first divided into different groups called **strata**.

A sample is then selected from each stratum.

Population

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Strata

Male Female

or

Faculty A Faculty B Faculty C

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Random Sample from Each Group

Example: Stratified Sampling

Suppose a university has:

- 60% undergraduate students
- 40% postgraduate students

The researcher wants a sample of 200 students.

A proportional sample could include:

- 120 undergraduate students
- 80 postgraduate students

Purpose

To ensure that important groups in the population are adequately represented.

4. Cluster Sampling

In **cluster sampling**, the population is divided into natural groups or clusters.

Instead of selecting individual units directly, the researcher randomly selects clusters.

Examples of Clusters

- Schools
- Wards
- Districts
- Villages
- Offices

Example: Cluster Sampling

Suppose a researcher wants to study teachers in Nepal.

Instead of selecting teachers from every school:

All Schools in Nepal

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Randomly Select Districts

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Select Schools

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Study Teachers

This approach can reduce the cost and difficulty of data collection.

Multistage Sampling

Multistage sampling involves selecting samples in several stages.

$$\textbf{Nepal}$$
$$\Downarrow$$
$$\textbf{Province}$$
$$\Downarrow$$
$$\textbf{District}$$
$$\Downarrow$$
$$\textbf{Municipality}$$
$$\Downarrow$$
$$\textbf{Ward}$$
$$\Downarrow$$
$$\textbf{Household}$$

Different sampling techniques may be used at different stages.

Non-Probability Sampling

In non-probability sampling, the probability of selecting each unit is **unknown**.

Selection may depend on:

- Accessibility
- Researcher's judgment
- Specific characteristics
- Availability of participants
- Recommendations from participants

Non-probability sampling is commonly used in qualitative research.

1. Convenience Sampling

In **convenience sampling**, participants are selected because they are easily available or accessible.

Example

A researcher surveys students who are available on campus during a particular day.

Advantages:

- Easy
- Quick
- Low cost

Limitation: High possibility of selection bias.

2. Purposive Sampling

In **purposive sampling**, participants are selected intentionally because they possess relevant knowledge, experience, or characteristics.

Example

To study the implementation of a government policy, the researcher may select:

- Policy makers
- Government officials
- Program managers
- Subject experts

Types of Purposive Sampling

Common types include:

- Typical case sampling
- Extreme or deviant case sampling
- Maximum variation sampling
- Homogeneous sampling
- Expert sampling
- Criterion sampling

Key Principle

Participants are selected because they can provide relevant and meaningful information for the research.

3. Quota Sampling

In **quota sampling**, the researcher selects participants until a predetermined number or quota is reached for particular categories.

Example

A researcher plans to interview:

- 50 male participants
- 50 female participants

Participants may be selected based on availability until each quota is completed.

Unlike stratified random sampling, selection within each group is not necessarily random.

4. Snowball Sampling

In **snowball sampling**, initial participants help the researcher identify or recruit other potential participants.

Participant 1

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Participant 2

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Participant 3

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Participant 4

It is particularly useful for studying:

- Hidden populations
- Difficult-to-reach groups
- Specialized networks

Probability vs. Non-Probability Sampling

Probability Sampling	Non-Probability Sampling
Known probability of selection	Unknown probability of selection
Random selection	Non-random selection
Useful for statistical generalization	Useful for in-depth understanding
Often used in quantitative research	Commonly used in qualitative research
Requires a sampling frame in many cases	May not require a complete sampling frame

How to Select a Sampling Technique

The selection of a sampling technique depends on:

- Research objectives
- Research questions
- Type of research
- Population characteristics
- Availability of a sampling frame
- Required level of generalization
- Time and financial resources
- Accessibility of participants

Matching Research and Sampling Techniques

Research Situation	Possible Sampling Technique
Large population with complete list	Simple Random Sampling
Population with different subgroups	Stratified Sampling
Geographically dispersed population	Cluster or Multistage Sampling
Easy access to participants	Convenience Sampling
Participants with specialized knowledge	Purposive Sampling
Difficult-to-reach population	Snowball Sampling

Sampling Errors and Bias

A sample may not perfectly represent the population.

Common problems include:

- Sampling error
- Selection bias
- Non-response bias
- Undercoverage
- Inappropriate sampling frame
- Inadequate sample size

Important

A good sampling technique helps reduce bias, but no sampling process is completely free from error.

Sampling in a Research Proposal

The sampling section of a research proposal should clearly explain:

1. Target population
2. Sampling unit
3. Sampling frame, where applicable
4. Sample size
5. Sampling technique
6. Procedure for selecting participants
7. Justification for the selected technique

Example of a Sampling Statement

Example

The study will use a **stratified random sampling technique** to select respondents from different categories of public service users. The population will first be divided into relevant groups, and a proportionate sample will be randomly selected from each group to ensure adequate representation.

The sampling technique should always be connected to the research objectives and design.

Overall Sampling Process

Define the Target Population

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Identify the Sampling Frame

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Determine Sample Size

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Select Sampling Technique

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Select the Sample

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Collect Data

Key Takeaway

Sampling is the process of selecting a part of the population to study the whole.

Population

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Sampling Frame

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Sampling Technique

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Sample

The appropriate sampling technique depends on the research problem, objectives, population, and research design.